

Marcus Thompson

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EXPERIENCE

Senior Software Engineer Spotify *iOS Client Platform* Feb 2020 – Present

- Redesigned audio playback engine reducing cold-start time from 3.2s to 0.8s, improving Day-1 retention by 6% across 180M mobile users
- Optimized memory footprint of offline sync module by 40% through custom memory-mapped file handling, eliminating OOM crashes on 3GB RAM devices
- Architected on-device listening history analysis using differential privacy, enabling personalization without transmitting raw user data to servers
- Leveraged Apple Neural Engine via Core ML for on-device audio fingerprinting, reducing server costs by 60% while improving accuracy to 99.2%
- Partnered with design team to implement fluid animations using Metal shaders, achieving 120fps on ProMotion displays while maintaining battery efficiency

Software Engineer Airbnb *Mobile Experience Team* Jul 2017 – Jan 2020

- Owned AR room measurement feature from prototype to launch, used by 2M+ hosts increasing booking conversion by 8%
- Reduced app binary size by 25% (18MB) through dead code elimination and asset optimization, improving install rate by 4% in emerging markets
- Led accessibility initiative achieving WCAG 2.1 AA compliance across 40+ screens, earning App Store editorial feature
- Implemented UI testing framework covering 85% of user flows, reducing regression bugs by 70%

iOS Developer Lyft *Rider App* Jun 2015 – Jun 2017

- Built custom map rendering layer using MapKit and Metal, achieving 60fps panning with real-time driver updates for 30M monthly riders
- Implemented offline-first architecture enabling ride requests in low-connectivity areas, recovering 3% of failed bookings

EDUCATION

B.S. Computer Science University of Michigan 2015
Minor: Human-Computer Interaction GPA: 3.8/4.0

PUBLISHED APPS & PROJECTS

Tempo (App Store, 50K+ downloads) – Minimal music practice timer with haptic metronome. SwiftUI, Core Haptics. Featured in “Apps We Love.”

FluidTransitions (Open Source) – Custom UIKit transition library with physics-based animations. 2.4K GitHub stars.

TECHNICAL SKILLS

Languages: Swift, Objective-C, C++, Python, Metal Shading Language

Frameworks: UIKit, SwiftUI, Core ML, ARKit, Core Animation, AVFoundation, Combine

Performance: Instruments, Metal, LLVM profiling, memory optimization, battery profiling

Practices: Accessibility (VoiceOver), Localization, Privacy by Design, Xcode Cloud